

Data Interpretation Exercise

Data Interpretation Assessment (R Programming)

Topics:

- Mapping Data onto Aesthetics
- Coordinate Systems and Axes
- Color Scales

Instructions: Use the given dataset to write an appropriate R program, generate the required visualization, and answer the questions.

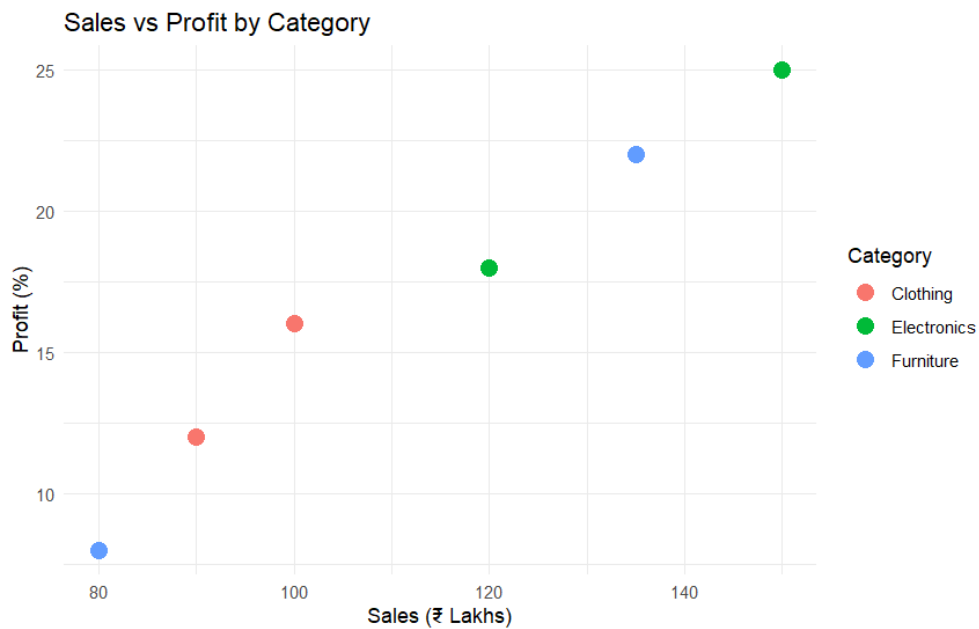
Question 1: Mapping Data onto Aesthetics

Dataset

Product	Sales (₹ Lakhs)	Profit (%)	Category
P1	120	18	Electronics
P2	90	12	Clothing
P3	150	25	Electronics
P4	80	8	Furniture
P5	135	22	Furniture
P6	100	16	Clothing

Questions

```
1 install.packages("ggplot2")
2 library(ggplot2)
3 data <- data.frame(
4   Product = c("P1", "P2", "P3", "P4", "P5", "P6"),
5   Sales = c(120, 90, 150, 80, 135, 100),
6   Profit = c(18, 12, 25, 8, 22, 16),
7   Category = c("Electronics", "Clothing", "Electronics", "Furniture", "Furniture", "Clothing")
8 )
9 ggplot(data, aes(x = Sales, y = Profit, color = Category)) +
10   geom_point(size = 4) +
11   labs(
12     title = "Sales vs Profit by Category",
13     x = "Sales (₹ Lakhs)",
14     y = "Profit (%)",
15     color = "Category"
16   ) +
17   theme_minimal()
```



2.

- **Category** is mapped to the **color** aesthetic to distinguish different product categories.
- **Sales** is mapped to the **size** aesthetic so that products with higher sales appear as larger points.

3. The scatter plot shows a **positive relationship** between Sales and Profit. Products with higher sales generally have higher profit percentages, although the relationship is not perfectly linear.

4. Sales (₹ Lakhs)- Continuous scale

Profit (%)-Continuous scale

Category- Discrete scale

5. A bubble chart is a better alternative because it can display Sales on the x-axis, Profit on the y-axis, Category by color, and Sales (or Profit) by bubble size, making it easier to compare all variables in a single visualization.

Question 2: Coordinate Systems (Cartesian and Polar)

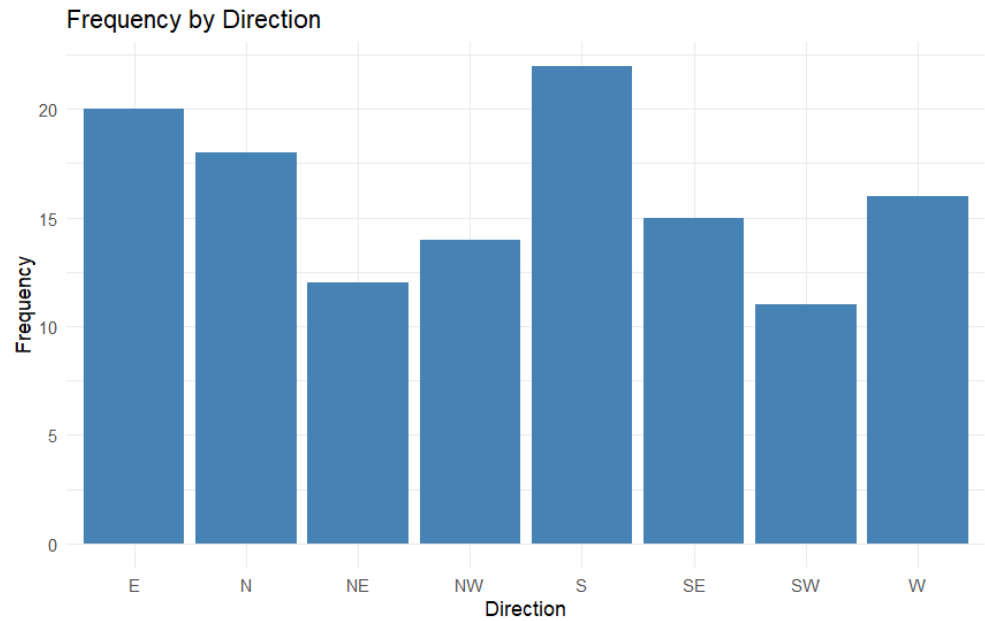
Dataset

Direction	Frequency
N	18
NE	12
E	20
SE	15
S	22
SW	11
W	16
NW	14

Questions

1.

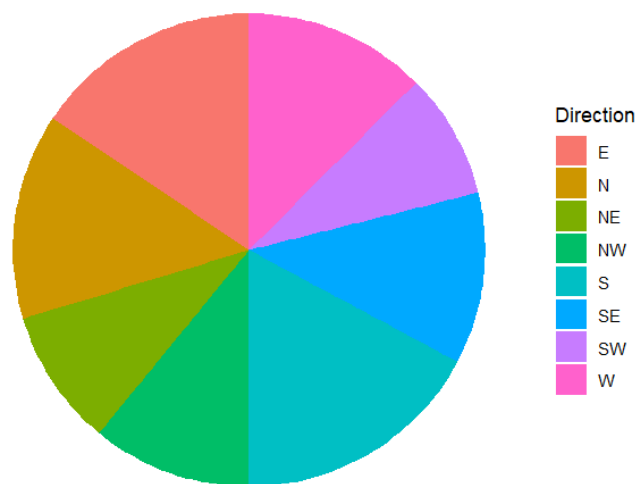
```
1 install.packages("ggplot2")
2 library(ggplot2)
3 data <- data.frame(
4   Direction = c("N", "NE", "E", "SE", "S", "SW", "W", "NW"),
5   Frequency = c(18, 12, 20, 15, 22, 11, 16, 14)
6 )
7 ggplot(data, aes(x = Direction, y = Frequency)) +
8   geom_bar(stat = "identity", fill = "steelblue") +
9   labs(
10     title = "Frequency by Direction",
11     x = "Direction",
12     y = "Frequency"
13   ) +
14   theme_minimal()
```



2.

```
1 install.packages("ggplot2")
2 library(ggplot2)
3 data <- data.frame(
4   Direction = c("N", "NE", "E", "SE", "S", "SW", "W", "NW"),
5   Frequency = c(18, 12, 20, 15, 22, 11, 16, 14)
6 )
7 ggplot(data, aes(x = "", y = Frequency, fill = Direction)) +
8   geom_bar(stat = "identity", width = 1) +
9   coord_polar(theta = "y") +
10  labs(
11    title = "Direction Frequency (Polar Coordinate System)",
12    fill = "Direction"
13  ) +
14  theme_void()
```

Direction Frequency (Polar Coordinate System)



3. The Cartesian (bar chart) is more effective for comparing frequencies accurately, while the Polar (pie chart) is better for showing the proportion of each direction. Overall, the Cartesian chart provides clearer comparisons.
4. The Polar coordinate system is more appropriate for directional data because it naturally represents directions as angles around a central point, making directional patterns easier to understand.
5. The Polar coordinate system better reflects the circular nature of directional data, making patterns across directions easier to interpret, while the Cartesian chart is better for comparing exact frequency values.

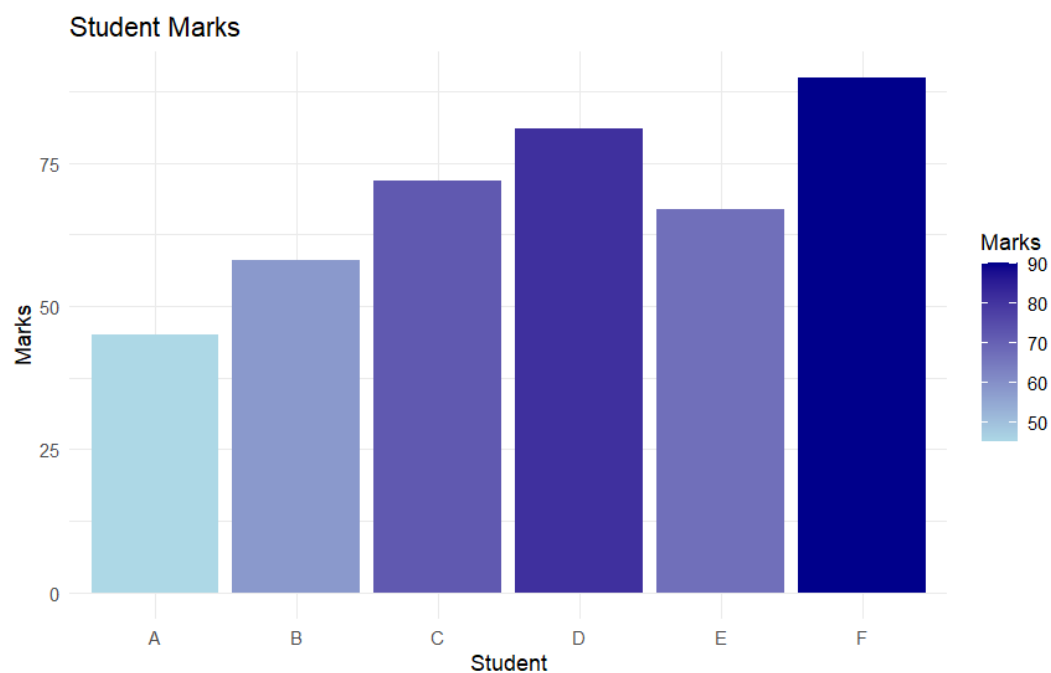
Question 3: Color Scales for Data Representation

Dataset

Student	Marks
A	45
B	58
C	72
D	81
E	67
F	90

Questions

```
1. 1 install.packages("ggplot2")
2   library(ggplot2)
3   data <- data.frame(
4     Student = c("A", "B", "C", "D", "E", "F"),
5     Marks = c(45, 58, 72, 81, 67, 90)
6   )
7   ggplot(data, aes(x = Student, y = Marks, fill = Marks)) +
8     geom_bar(stat = "identity") +
9     scale_fill_gradient(low = "lightblue", high = "darkblue") +
10    labs(
11      title = "Student Marks",
12      x = "Student",
13      y = "Marks",
14      fill = "Marks"
15    ) +
16    theme_minimal()
```



- Student F has the highest marks (90), and Student A has the lowest marks (45).
- A sequential color scale is appropriate because the marks are continuous numerical values. It uses lighter-to-darker shades to clearly represent lower-to-higher marks.
- The Viridis color palette is suitable for color-blind users because it is perceptually uniform and easy to distinguish.
- A sequential color scale represents ordered numerical data using a gradual change in color intensity, while a categorical color scale uses distinct colors to represent different categories without any order.

Question 4: Highlighting with Color

Dataset

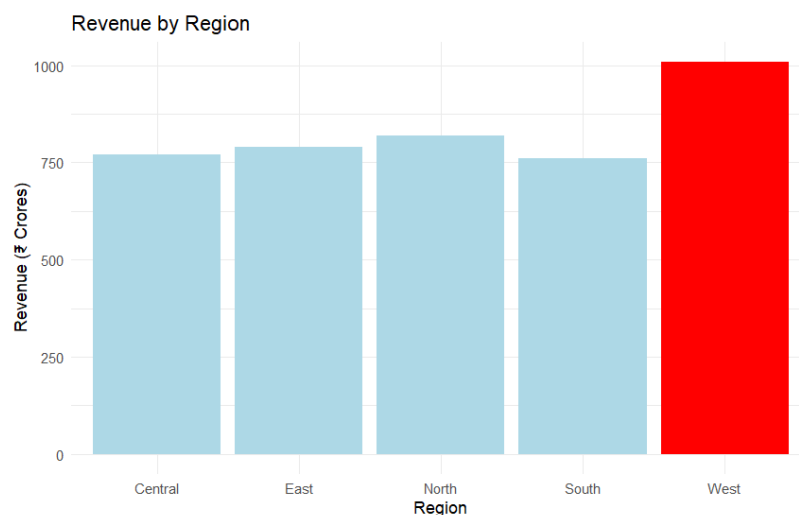
Region	Revenue (₹ Crores)
North	820
South	760
East	790
West	1010
Central	770

Questions

- ```

1 library(ggplot2)
2 data <- data.frame(
3 Region = c("North", "South", "East", "West", "Central"),
4 Revenue = c(820, 760, 790, 1010, 770)
5)
6 data$Highlight <- ifelse(data$Revenue == max(data$Revenue), "Highest", "Other")
7 ggplot(data, aes(x = Region, y = Revenue, fill = Highlight)) +
8 geom_bar(stat = "identity") +
9 scale_fill_manual(values = c("Highest" = "red", "Other" = "lightblue")) +
10 labs(
11 title = "Revenue by Region",
12 x = "Region",
13 y = "Revenue (₹ Crores)"
14) +
15 theme_minimal() +
16 theme(legend.position = "none")

```



2. Highlighting draws attention to the most important data point, making it easier to identify and compare. It improves readability by emphasizing key information, such as the highest revenue region.
3. The West region is the highest-performing region with a revenue of. ₹1010 crores
4. Excessive use of colors can make a chart confusing and distract viewers from important information. Using too many colors reduces clarity and makes comparisons more difficult.
5. Using labels or annotations is an effective alternative highlighting technique to emphasize important data points without relying on color.

Question 5: Integrated Visualization

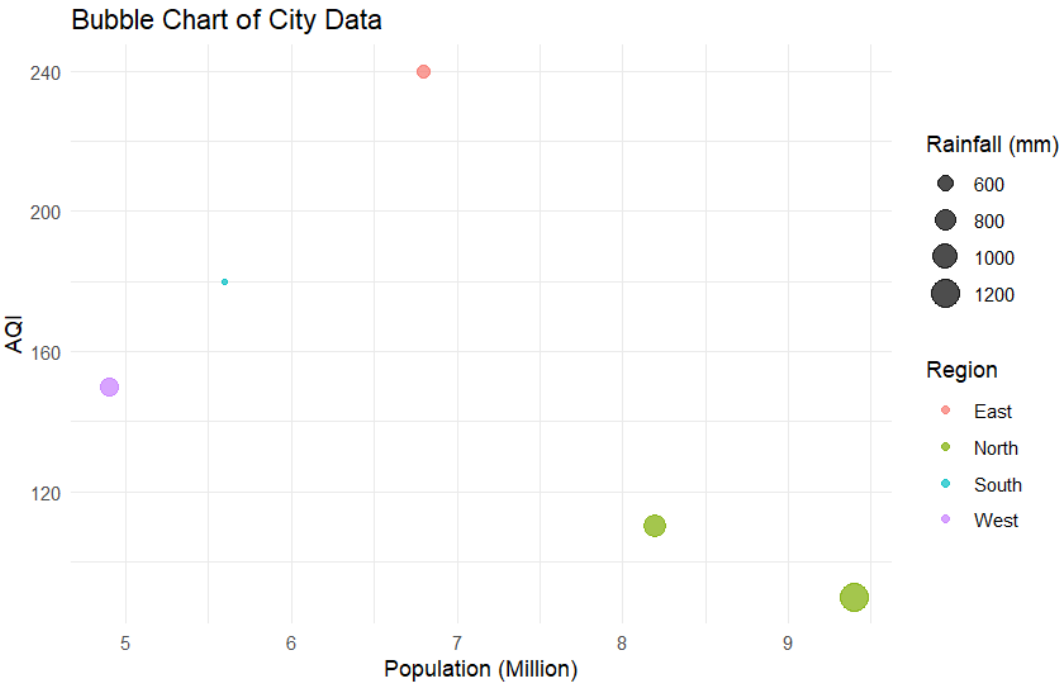
Dataset

| City | Population (Million) | AQI | Rainfall (mm) | Region |
|------|----------------------|-----|---------------|--------|
| A    | 8.2                  | 110 | 850           | North  |
| B    | 5.6                  | 180 | 430           | South  |
| C    | 9.4                  | 90  | 1200          | North  |
| D    | 6.8                  | 240 | 520           | East   |
| E    | 4.9                  | 150 | 680           | West   |

Questions

1.

```
1 library(ggplot2)
2 data <- data.frame(
3 City = c("A", "B", "C", "D", "E"),
4 Population = c(8.2, 5.6, 9.4, 6.8, 4.9),
5 AQI = c(110, 180, 90, 240, 150),
6 Rainfall = c(850, 430, 1200, 520, 680),
7 Region = c("North", "South", "North", "East", "West")
8)
9 ggplot(data, aes(x = Population, y = AQI,
10 size = Rainfall,
11 color = Region)) +
12 geom_point(alpha = 0.7) +
13 labs(
14 title = "Bubble Chart of City Data",
15 x = "Population (Million)",
16 y = "AQI",
17 size = "Rainfall (mm)",
18 color = "Region"
19) +
20 theme_minimal()
```





2. Population → X-axis, AQI → Y-axis, Rainfall → Bubble size, Region → Color.  
This aesthetics clearly display relationships, magnitude, and regional categories in a single visualization.
3. The dataset shows no clear relationship between Population and AQI, as cities with higher populations do not consistently have higher AQI values.
4. **Continuous variables:** Population, AQI, and Rainfall.  
**Categorical variables:** City and Region.
5. Add data labels to each bubble so the values are easier to identify and compare.